

AI Architecture Document (AAD)

IntelliPlant AI

AI-Powered Industrial Knowledge Intelligence Platform

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1. Introduction

IntelliPlant AI transforms heterogeneous industrial documents into a unified operational brain through:

- OCR
- Entity Extraction
- Knowledge Graphs
- GraphRAG
- Hybrid Search
- Multi-Agent Systems

- LLM Reasoning

The architecture is designed to provide explainable, reliable, and context-aware industrial intelligence.

2. AI Objectives

The AI system should:

Knowledge Discovery

Find hidden relationships.

Conversational Assistance

Answer engineering questions.

Root Cause Analysis

Analyze failures and incidents.

Compliance Intelligence

Detect regulatory gaps.

Lessons Learned Intelligence

Identify recurring patterns.

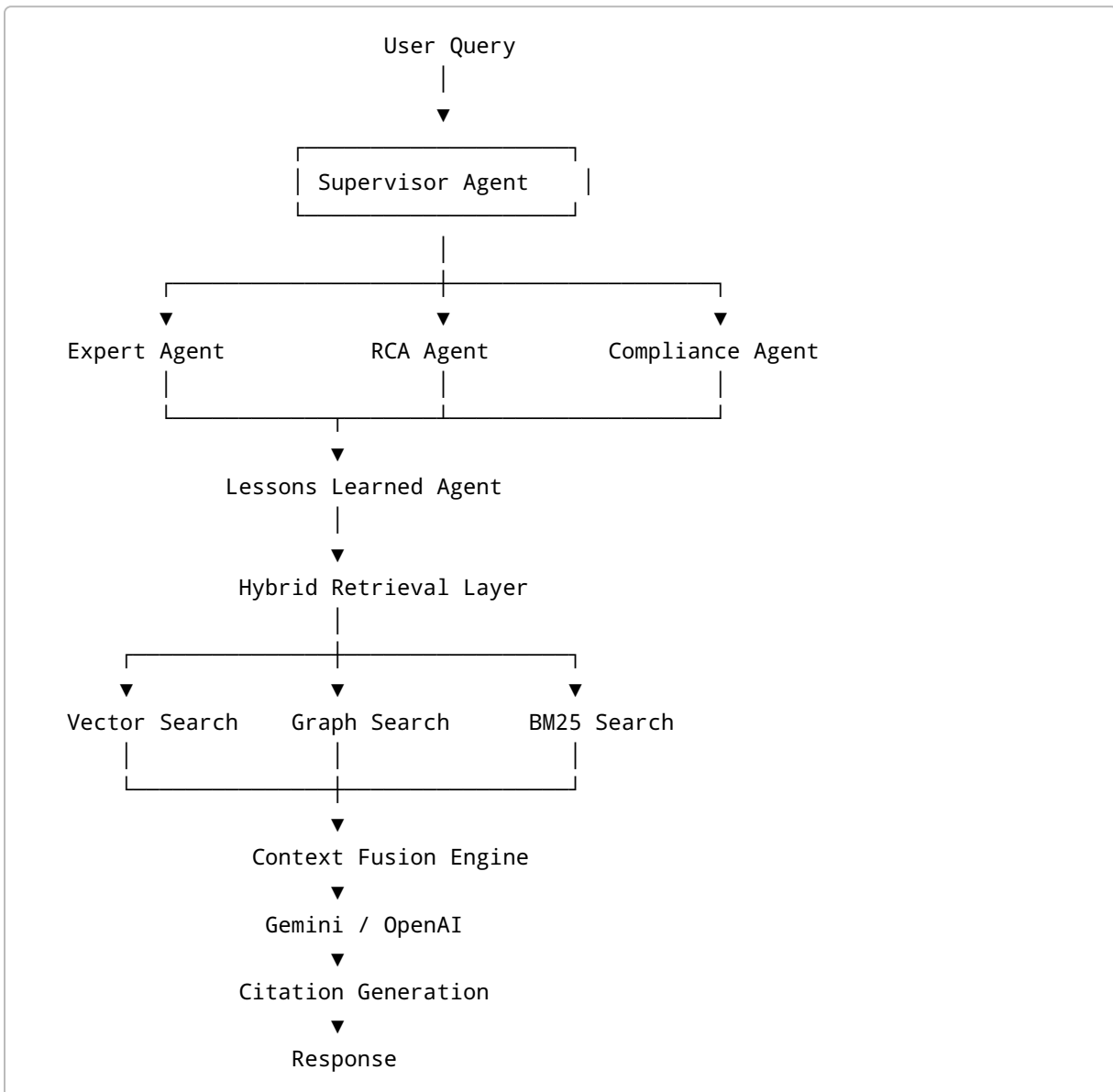
Explainability

Provide source citations.

Hallucination Prevention

Ground all responses on evidence.

3. Overall AI Architecture



4. AI Layers

The AI system consists of:

Layer 1

Document Intelligence

Layer 2

Knowledge Layer

Layer 3

Retrieval Layer

Layer 4

Multi-Agent Layer

Layer 5

Reasoning Layer

Layer 6

Memory Layer

Layer 7

Explainability Layer

5. Document Intelligence Layer

Purpose:

Convert raw documents into structured knowledge.

Inputs

PDF

DOCX

Images

Scanned Manuals

Inspection Reports

Emails

P&IDs

Components

OCR Engine

Metadata Extractor

Entity Extractor

Chunk Generator

Embedding Engine

Output

Structured JSON

Example

```
{  
  "equipment": "Pump P101",  
  "failure": "Bearing Failure",  
  "date": "2025-10-01"  
}
```

6. Knowledge Layer

Purpose

Create industrial knowledge relationships.

Knowledge Graph

Technology

Neo4j

Nodes

Equipment

Incident

Manual

Inspection

Failure

Operator

Regulation

WorkOrder

Relationships

CONNECTED_TO

FAILED_IN

CAUSES

SIMILAR_TO

REQUIRES

DESCRIBED_BY

Example

```
Pump P101
  ↓
FAILED_IN
  ↓
Bearing Failure
  ↓
CAUSES
  ↓
Overheating
```

7. Retrieval Layer

Hybrid Retrieval combines:

Vector Retrieval

Qdrant

Semantic similarity

Graph Retrieval

Neo4j

Relationship traversal

BM25 Retrieval

Keyword search

Retrieval Flow

```
User Query
  ↓
Query Rewriter
  ↓
-----
Vector Search
Graph Search
```

BM25 Search

↓

Context Fusion

8. Multi-Agent Layer

LangGraph orchestrates multiple agents.

Supervisor Agent

Responsibilities

Intent classification

Agent routing

Context management

Response aggregation

Expert Copilot Agent

Handles

SOP retrieval

Engineering Q&A

Manual lookup

RCA Agent

Handles

Failure analysis

Historical incidents

Maintenance recommendations

Compliance Agent

Handles

Factory Act

OISD

ISO Standards

PESO

Lessons Agent

Handles

Near misses

Incident patterns

Recommendations

9. Reasoning Layer

LLM performs:

Chain-of-Thought

Stepwise reasoning

Tool Calling

Invoke retrieval tools

Context Synthesis

Combine multiple sources

Explanation Generation

Generate human-readable answers

Example

Query

Why Pump P101 failed?

↓

Retrieve incidents

↓

Retrieve manual

↓

Retrieve similar failures

↓

Analyze causes

↓

Generate explanation

10. LLM Layer

Primary Model

Gemini 2.5 Pro

Fallback Model

GPT-4o

Future Models

Claude

Llama 4

DeepSeek

Responsibilities

Question answering

Reasoning

Summarization

Recommendations

Citation generation

11. Memory Architecture

Short-Term Memory

Technology

Redis

Stores

Recent conversations

Session state

Context

Long-Term Memory

Technology

Neo4j + PostgreSQL

Stores

Historical incidents

Documents

Relationships

Semantic Memory

Technology

Qdrant

Stores

Embeddings

Chunks

Vectors

12. Context Management

Conversation History



Memory Retrieval



Hybrid Search



Context Compression



LLM

13. Confidence Scoring

Final Confidence Score

```
Confidence =  
0.4 × Retrieval Score +  
0.3 × Citation Score +  
0.3 × Graph Evidence Score
```

Range

0-1

Example

```
{  
  "confidence":0.92  
}
```

14. Hallucination Prevention

Grounded Responses

Only retrieved context is used.

Source Citations

Every answer includes references.

Confidence Threshold

Below 0.7:

Return

"Insufficient evidence."

Multi-source Verification

Cross-check:

Vector search

Graph search

Keyword search

15. Explainability Layer

Every response contains

Answer

Human-readable explanation

Sources

Manuals

Incidents

Reports

Related Entities

Equipment

Failures

Operators

Confidence Score

0-1

Example

```
{  
  "answer": "",  
  "sources": [  
    "manual.pdf",  
    "incident_2024.pdf"  
  ]  
}
```

```
],  
  "confidence":0.91  
}
```

16. Embedding Architecture

Model

BGE-Large

Dimension

1024

Chunk Size

1024

Overlap

200

Pipeline

```
graph TD  
  Document --> Chunking  
  Chunking --> Embedding_Model[Embedding Model]  
  Embedding_Model --> Qdrant
```

17. AI Workflow

```
graph TD  
  Document_Upload[Document Upload] --> OCR  
  OCR --> 
```

Entity Extraction



Knowledge Graph



Embeddings



Qdrant



Ready

Query Flow

Question



Supervisor Agent



Agent Selection



Hybrid Retrieval



LLM



Citation Engine



Response

18. AI Metrics

Metric	Target
Citation Accuracy	>95%
Hallucination Rate	<5%
Retrieval Precision	>90%
Graph Completeness	>85%
Answer Accuracy	>90%
Query Latency	<5 sec
Confidence Calibration	>90%

19. Failure Handling

LLM Failure

Fallback model

Qdrant Failure

BM25 fallback

Neo4j Failure

Vector search only

OCR Failure

Retry queue

Agent Failure

Supervisor reroutes task

20. Future AI Enhancements

GraphRAG 2.0

Hierarchical Agents

Self-Learning Memory

Digital Twin AI

Predictive Maintenance Models

Multimodal LLMs

Voice Copilot

Real-Time IoT Intelligence

Autonomous Agents

Conclusion

The IntelliPlant AI Architecture combines Document Intelligence, Knowledge Graphs, Hybrid Retrieval, GraphRAG, Multi-Agent Systems, and Explainable LLMs to create an Industrial Operations Brain capable of delivering trustworthy and context-aware intelligence to maintenance engineers, quality teams, and plant managers.